ADAPTIVE MULTI-JUNCTION TRAFFIC GRID CONTROLLER (AMTGC)

PROJECT REPORT

Abstract

The AMTGC project focuses on improving traffic management at two connected junctions. It uses Verilog HDL to control traffic signals, coordinate green lights, manage pedestrian requests, and handle emergency situations. The design was tested using ModelSim under different traffic conditions. The simulation showed successful results, including an 8-cycle green-wave delay and fair pedestrian service at both junctions. Overall, this project demonstrates how a simple digital system can help make traffic control more organised and efficient.

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1. PROJECT OVERVIEW

1.1 Introduction

Traffic congestion is something we experience regularly, especially near busy road junctions. When several roads meet, managing the movement of vehicles becomes difficult. The situation becomes even more challenging when traffic density changes, pedestrians need to cross the road, or an emergency vehicle needs to pass. To address these problems, this project focuses on developing an Adaptive Multi-Junction Traffic Grid Controller (AMTGC) using Verilog HDL. The system is designed to control two connected four-way traffic junctions, called Junction A and Junction B. The main idea behind this project is to make the operation of both junctions more coordinated instead of treating them as two completely separate traffic signals. Along with normal traffic control, the system also considers pedestrian requests and emergency situations. The complete design was developed using separate Verilog modules and tested through simulation in ModelSim.

1.2 Objectives

The main purpose of this project is to design a traffic controller that can manage two connected junctions in an organized manner.

The objectives include:

- Controlling traffic movement in the North-South and East-West directions.
- Coordinating the signals at both junctions using green-wave timing.
- Handling pedestrian requests from both junctions through a common arbiter.
- Giving priority to emergency conditions.
- Testing the design under different traffic conditions.
- Checking the performance of the system using ModelSim waveforms.

1.3 Scope of the Project

This project focuses on the digital design and simulation of a two-junction traffic control system. The complete system is implemented using Verilog HDL and verified through testbenches in ModelSim. The project does not involve a physical traffic signal setup. Instead, the working of the controller is studied through simulation results and waveform observations.

2. ARCHITECTURE

2.1 System Architecture

The AMTGC project consists of two connected four-way traffic junctions. Each junction has traffic signals that control vehicles moving in the North-South and East-West directions. While both junctions have their own traffic controllers, certain functions are handled centrally so that the overall system works in a coordinated manner. The design is divided into different modules. Each module has a specific responsibility, which makes the overall design easier to understand and maintain.

The major modules used in the project are:

- Generic Timer
- Junction Controller
- Pedestrian Arbiter
- Green-Wave Coordinator
- Emergency Override
- AMTGC Top Module

2.2 Working of the System

During normal operation, the junction controllers manage the traffic signals according to their programmed sequence. The signals change between different phases to allow vehicles from different directions to move safely. The green-wave coordinator helps maintain a timing relationship between the two junctions. This allows the North-South green signal at Junction B to start after a defined delay from Junction A. Pedestrian requests from both junctions are handled by a shared pedestrian arbiter. Using one common arbiter helps manage the requests centrally rather than having separate arbitration units for each junction. The emergency override is included to handle situations where an emergency vehicle needs priority. When the emergency input is activated, both junctions are forced into an all-red condition. All these modules are connected through the AMTGC top module, which brings the complete design together.

2.3 Input and Output Signals

The system receives different inputs such as clock, reset, traffic-density information, pedestrian requests, and emergency signals. Based on these inputs, the controller generates the required traffic light outputs for both junctions and manages pedestrian service.

3. RTL DESIGN

3.1 Introduction

The complete traffic controller was written using Verilog HDL. Instead of writing the entire design as one large program, it was divided into smaller modules. This approach made it easier to work on individual functions, test them separately, and combine them into the final system.

3.2 Description of the Modules

Generic Timer

The generic timer is used to manage the timing required by the traffic controller. Traffic signals need to remain active for a particular duration before changing to the next phase. This module supports the timing operations required by the design.

Junction Controller

The junction controller is responsible for managing the traffic signals at each junction. It controls the movement of vehicles in the North-South and East-West directions. It follows the programmed traffic sequence and changes the signal phases based on the timing and control conditions.

Pedestrian Arbiter

One important part of this project is pedestrian request handling. Since requests can come from both Junction A and Junction B, a shared pedestrian arbiter is used. It manages the requests from both junctions and coordinates their service. This avoids the need for two separate arbitration units.

Green-Wave Coordinator

The green-wave coordinator is used to maintain coordination between the two junctions. The main purpose of this module is to ensure that the North-South green signal at Junction B starts after the required delay from Junction A. This timing relationship is useful for maintaining coordinated traffic movement.

Emergency Override

Emergency situations need immediate attention in a traffic control system. For this reason, an emergency override module is included. When the emergency input is activated, the system overrides normal traffic operation and forces both junctions into an all-red condition.

AMTGC Top Module

The AMTGC top module connects all the individual modules and makes them work together as one complete system. It handles the connections between the junction controllers, timer, pedestrian arbiter, green-wave coordinator, and emergency override.

4. FSM DESIGN

4.1 Introduction to Finite State Machines

A Finite State Machine (FSM) is a method used in digital systems to control operations through a set of predefined states. In a traffic signal controller, the signals cannot change randomly. They need to follow a proper sequence so that vehicles from different directions can move safely. For this reason, FSM-based control is used in the traffic controller design.

4.2 Traffic Signal Operation

The traffic controller works through different traffic phases. During the North-South green phase, vehicles travelling in that direction are allowed to move. After the required duration, the signal changes to yellow to indicate that the phase is ending. The controller then allows traffic in the East-West direction to move through its green phase, followed by the corresponding yellow phase. This sequence continues during normal operation.

4.3 State Transitions

The controller moves from one state to another depending on the clock, timing conditions, and control inputs. The timing mechanism helps determine when a particular phase should end and when the next phase should begin. Emergency conditions are given priority over normal traffic operation. When an emergency occurs, the normal sequence is overridden and both junctions are forced into an all-red condition. This organized state-based approach makes the traffic control operation easier to manage and verify.

5. VERIFICATION METHODOLOGY

5.1 Introduction

After completing the RTL design, it was important to check whether all the modules were working as expected. For this purpose, Verilog testbenches were used to provide different input conditions to the design. The outputs were then observed using ModelSim. Verification helped identify whether the traffic controller behaved correctly under normal as well as special operating conditions.

5.2 Simulation Setup

ModelSim Intel FPGA Starter Edition 2020.1 was used for compiling and simulating the Verilog files. The testbenches were used to apply different input combinations and observe the responses of the controller. The waveform window was particularly useful because it allowed the signal transitions to be studied over time.

5.3 Test Scenarios

The complete design was tested under several conditions.

Traffic-density testing: Different combinations of traffic density at Junction A and Junction B were tested to check the controller under different traffic conditions.

Green-wave testing: The timing difference between the North-South green signals of both junctions was measured to verify the required coordination.

Pedestrian testing: Requests from both junctions were tested to check whether the shared pedestrian arbiter could provide service.

Emergency testing: Emergency conditions were activated during different normal traffic phases to check whether the emergency override worked.

Reset testing: Reset conditions were applied to observe the behaviour of the system during initialization and reset operations.

Rapid emergency testing: The emergency input was changed repeatedly to check the controller's response to rapid input changes.

Random testing: A deterministic pseudo-random test sequence was applied for 1000 cycles to check the design under different combinations of inputs.

6. SIMULATION RESULTS

6.1 Simulation Overview

The complete AMTGC design was simulated in ModelSim using the full verification testbench. During simulation, different traffic-density combinations, pedestrian requests, emergency conditions, reset operations, and random input sequences were tested. The simulation results were observed through the transcript window and waveform viewer.

6.2 Green-Wave Results

One of the important functions of this project is the coordination between Junction A and Junction B. During verification, all eight tested traffic-density combinations passed the green-wave check. The measured timing difference between the North-South green signals of both junctions was eight clock cycles. This showed that the required timing relationship was maintained in the tested cases.

6.3 Pedestrian Results

The shared pedestrian arbiter was tested using requests from both junctions. The simulation reported 120 pedestrian services for Junction A and 120 pedestrian services for Junction B. These results show that pedestrian service was exercised for both junctions during the verification process.

6.4 Emergency and Reset Results

Emergency conditions were tested during each ordinary traffic phase or state. The simulation also included rapid emergency input changes and active reset conditions. These tests helped check the behaviour of the controller when normal operation was interrupted.

6.5 Overall Simulation Result

The full verification testbench completed successfully.

The final transcript displayed:

FULL_REGRESSION_PASS

The simulation reached its completion point at 88,580 ns.

The successful regression result indicates that all the checks included in the full testbench passed during this simulation run.

Transcript

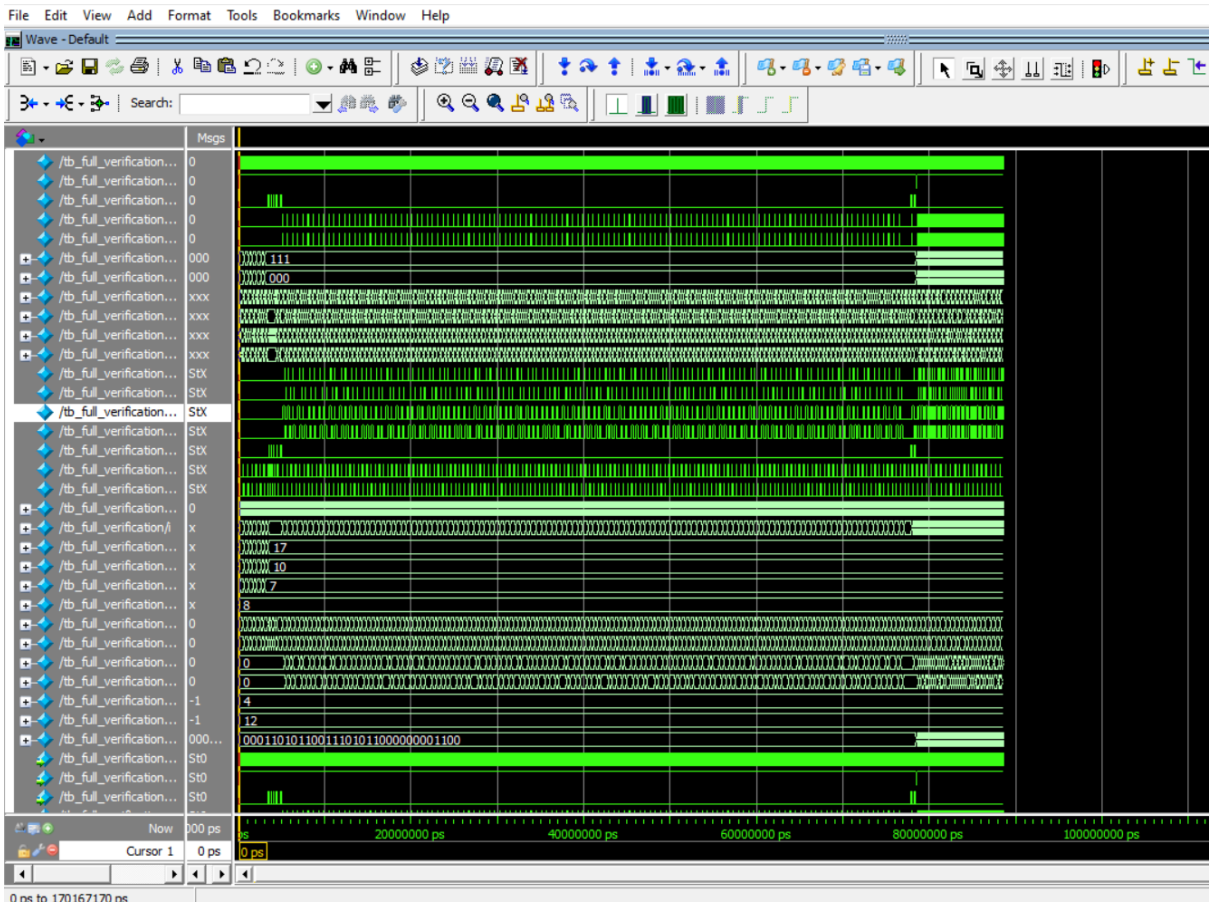
File Edit View Bookmarks Window Help

```
# Reading pref.tcl
ModelSim> cd C:/Users/Lenovo/Downloads/AMTGC_project/AMTGC
ModelSim> vlib work
# ** Warning: (vlib-34) Library already exists at "work".
ModelSim> vsim work.tb_full_verification
# vsim work.tb_full_verification
# Start time: 22:33:14 on Nov 05,2026
# Loading work.tb_full_verification
# Loading work.amtgc_top
# Loading work.emergency_override
# Loading work.ped_arbiter
# Loading work.green_wave_coordinator
# Loading work.junction_controller
# Loading work.generic_timer
VSIM 4> add wave -r sim:/tb_full_verification/*
VSIM 5> view wave
# .main_pane.wave.interior.cs.body.pw.wf
VSIM 6> run -all
# PASS: density combo A=0 B=7, measured green-wave offset=8 cycles
# PASS: density combo A=1 B=6, measured green-wave offset=8 cycles
# PASS: density combo A=2 B=5, measured green-wave offset=8 cycles
# PASS: density combo A=3 B=4, measured green-wave offset=8 cycles
# PASS: density combo A=4 B=3, measured green-wave offset=8 cycles
# PASS: density combo A=5 B=2, measured green-wave offset=8 cycles
# PASS: density combo A=6 B=1, measured green-wave offset=8 cycles
# PASS: density combo A=7 B=0, measured green-wave offset=8 cycles
# PASS: emergency exercised during each ordinary traffic phase/state
# PASS: pedestrian service counts A=120 B=120
# PASS: rapid emergency toggles and active reset exercised
# PASS: 1000-cycle deterministic pseudo-random stimulus completed
# FULL_REGRESSION_PASS (run in ModelSim and archive its actual transcript/waveform before submission)
# ** Note: $finish    : tb/tb_full_verification.v(170)
#    Time: 88580 ns  Iteration: 1  Instance: /tb_full_verification
# 1
# Break in Module tb_full_verification at tb/tb_full_verification.v line 170
VSIM 7> wave zoom full
# {0 ps} {93009 ns}

VSIM 8>
```

Wave

File Edit View Add Format Tools Bookmarks Window Help



7. CHALLENGES FACED

While developing this project, a few challenges were encountered during the design and simulation stages. One of the main challenges was coordinating two traffic junctions without disturbing their normal signal sequence. Both junctions needed to operate in an organized manner while maintaining the required green-wave timing. Handling pedestrian requests from two different junctions was another important part of the project. Using a shared pedestrian arbiter required careful consideration of how requests from both sides would be managed. The emergency override also needed attention because emergency conditions had to take priority over normal traffic operation. There were also some difficulties during simulation in ModelSim. At one stage, the full regression testbench showed a timeout instead of completing successfully. After further work, a later complete simulation run finished successfully and displayed FULL_REGRESSION_PASS.

Working through these issues helped me understand the importance of testing individual modules, checking waveforms, and performing complete system verification.

8. CONCLUSION

The AMTGC project was developed to study how two connected traffic junctions can be controlled and coordinated using digital logic. The system was implemented using Verilog HDL and simulated in ModelSim. It combines normal traffic signal control with green-wave coordination, shared pedestrian request handling, and emergency override functionality. The verification results showed that all eight tested traffic-density combinations passed the green-wave check with an eight-clock-cycle offset. Pedestrian service was recorded for both junctions, and emergency and reset conditions were also exercised.

The complete regression testbench finished successfully and displayed FULL_REGRESSION_PASS.

Overall, this project gave me practical experience in writing modular Verilog code, understanding FSM-based control, connecting different RTL modules, creating testbenches, and analysing simulation waveforms. It also helped me understand how digital design concepts can be applied to a real-world problem such as traffic management.